Observability matrix and Kalman filter, from the start (first-order gain model)

1 The observability matrix

1.1

$$\mathbf{x} = \begin{bmatrix} \delta x_1 \\ \delta x_2 \\ \delta x_3 \\ a_x \end{bmatrix}, \qquad \mathbf{e} = \mathbf{x} - \hat{\mathbf{x}} = \begin{bmatrix} e_{\delta x_1} \\ e_{\delta x_2} \\ e_{\delta x_3} \\ e_{a_x} \end{bmatrix}, \qquad \mathbf{y} - \hat{\mathbf{y}} = \mathbf{H}\,\mathbf{e} + \mathbf{n}, \qquad \mathbf{H} = \begin{bmatrix} \mathbf{H}_1 \\ \mathbf{H}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

1.2

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k], \qquad \mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{e}[k_0+j] = \mathbf{\Phi}(k_0+j, k_0)\,\mathbf{e}[k_0]$$

$$\mathbf{\Phi}(k_0+j, k_0) = \mathbf{F}_e[k_0+j-1]\cdots\mathbf{F}_e[k_0], \qquad \mathbf{\Phi}(k_0, k_0) = \mathbf{I}$$

1.3

$$\begin{bmatrix} \mathbf{y}[k_0] - \hat{\mathbf{y}}[k_0] \\ \mathbf{y}[k_0+1] - \hat{\mathbf{y}}[k_0+1] \\ \vdots \\ \mathbf{y}[k_0+N] - \hat{\mathbf{y}}[k_0+N] \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{H} \\ \mathbf{H}\,\mathbf{\Phi}(k_0+1, k_0) \\ \vdots \\ \mathbf{H}\,\mathbf{\Phi}(k_0+N, k_0) \end{bmatrix}}_{\mathbf{O}(k_0)} \mathbf{e}[k_0] + \mathbf{n}$$

1.4

$$\mathbf{O}\,\mathbf{e}[k_0] = \mathbf{o}_{\delta x_1}\, e_{\delta x_1}[k_0] + \mathbf{o}_{\delta x_2}\, e_{\delta x_2}[k_0] + \mathbf{o}_{\delta x_3}\, e_{\delta x_3}[k_0] + \mathbf{o}_{a_x}\, e_{a_x}[k_0], \qquad \mathbf{o}_j = \mathbf{O}\,\hat{\mathbf{e}}_j$$

1.5

$$\mathbf{O}\,\mathbf{e} = \mathbf{O}\,\mathbf{e}' \;\Rightarrow\; \mathbf{e} = \mathbf{e}' \quad\Leftrightarrow\quad \operatorname{rank}\mathbf{O} = 4, \qquad\qquad \text{weight of column } j: \;\; \mathbf{o}_j^\top \mathbf{R}^{-1} \mathbf{o}_j \quad \text{(page 3)}$$

2 The a_x column in y_1 and y_2

2.1 $\mathbf{e}[k_0] = [\,0\;\;0\;\;0\;\;1\,]^\top$

$$\mathbf{e}[k_0] = [\,0,\;\;0,\;\;0,\;\;1\,]^\top$$
$$\mathbf{e}[k_0+1] = [\,0,\;\;0,\;\;-F_{dx}[k_0],\;\;1\,]^\top$$
$$\mathbf{e}[k_0+2] = [\,0,\;\;-F_{dx}[k_0],\;\;-\lambda_c F_{dx}[k_0] - F_{dx}[k_0+1],\;\;1\,]^\top$$
$$\mathbf{e}[k_0+3] = [\,-F_{dx}[k_0],\;\;-\lambda_c F_{dx}[k_0] - F_{dx}[k_0+1],\;\;-\lambda_c^2 F_{dx}[k_0] - \lambda_c F_{dx}[k_0+1] - F_{dx}[k_0+2],\;\;1\,]^\top$$

2.2

$$e_{a_x}[k_0+j] = 1$$
$$e_{\delta x_3}[k_0+j] = -\sum_{i=0}^{j-1} \lambda_c^{j-1-i}\, F_{dx}[k_0+i]$$
$$e_{\delta x_2}[k_0+j] = e_{\delta x_3}[k_0+j-1], \qquad e_{\delta x_1}[k_0+j] = e_{\delta x_3}[k_0+j-2]$$

2.3 $\mathbf{o}_{a_x}$, row by row

$$y_2[k_0+j] - \hat{y}_2[k_0+j] = \mathbf{H}_2\,\mathbf{e}[k_0+j] = 1 \qquad (j \geq 0)$$
$$y_1[k_0+j] - \hat{y}_1[k_0+j] = \mathbf{H}_1\,\mathbf{e}[k_0+j] = 0 \qquad (j = 0,1,2)$$
$$= -\sum_{i=0}^{j-3} \lambda_c^{j-3-i}\, F_{dx}[k_0+i] \quad (j \geq 3)$$

2.4 F_{dx} constant

$$y_1[k_0{+}j]{-}\hat{y}_1[k_0{+}j] = -F_{dx}\,\frac{1-\lambda_c^{j-2}}{1-\lambda_c} \xrightarrow{\;j\to\infty\;} -\frac{F_{dx}}{1-\lambda_c}, \qquad F_{dx} = \big(1{+}(1{-}\lambda_c)\,d\big)\,\frac{\Delta x_d}{\hat{a}_x}, \qquad \Delta x_d = v\,T_s$$

2.5 the first rows of $\mathbf{O}$ (F_{dx} constant)

$\mathbf{O} =$

	$e_{\delta x_1}$	$e_{\delta x_2}$	$e_{\delta x_3}$	e_{a_x}
$y_1[k_0]$	1	0	0	0
$y_2[k_0]$	0	0	0	1
$y_1[k_0+1]$	0	1	0	0
$y_2[k_0+1]$	0	0	0	1
$y_1[k_0+2]$	0	0	1	0
$y_2[k_0+2]$	0	0	0	1
$y_1[k_0+3]$	0	0	λ_c	$-F_{dx}$
$y_2[k_0+3]$	0	0	0	1
$y_1[k_0+4]$	0	0	λ_c^2	$-F_{dx}(1+\lambda_c)$
$y_2[k_0+4]$	0	0	0	1
$\vdots$				$\vdots$

2.6 hold ($\Delta x_d = 0 \Rightarrow F_{dx} = 0$)

$\mathbf{o}_{a_x} = [\,0,\;1,\;0,\;1,\;0,\;1,\;\ldots\,]^\top$ (y_1 rows 0, y_2 rows 1)

3 c-only: the model and the c column

3.1 $\hat{h}[k] = x_d[k] - \delta\hat{x}_3[k] - x_s, \quad M[k] = \Delta x_d[k] + (1-\lambda_c)\,\delta\hat{x}_3[k], \quad \nabla_d x_d[k] = x_d[k] - x_d[k-d]$

$$a'(\hat{h};\, c) = \frac{1-c}{\hat{h}^2}, \qquad \frac{\partial a'}{\partial c} = -\frac{1}{\hat{h}^2}, \qquad \frac{\partial a'}{\partial \hat{h}} = -\frac{2a'}{\hat{h}}$$

$$\hat{a}_x[k+1] = \hat{a}_x[k] + a'\big(\hat{h}[k];\, \hat{c}[k]\big)\, M[k], \qquad \hat{c}[k+1] = \hat{c}[k]$$

$$y_1[k] = \delta x_1[k] + n_x[k], \qquad y_2[k] = a_x[k-d] + n_a[k], \qquad \hat{y}_2[k] = \hat{a}_x[k] - a'[k]\,\nabla_d x_d[k]$$

3.2 $\mathbf{e} = [\, e_{\delta x_1} \;\; e_{\delta x_2} \;\; e_{\delta x_3} \;\; e_{a_x} \;\; e_c \,]^\top$; $\alpha := 1 + a' F_{dx}$, $\beta := \dfrac{M}{\hat{h}^2}$, $\gamma := \dfrac{\nabla_d x_d}{\hat{h}^2}$, $\eta := a'(1-\lambda_c) + \dfrac{2a'M}{\hat{h}}$, $\mu := \dfrac{2a'\,\nabla_d x_d}{\hat{h}}$, $q := F_{dx}$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -q & 0 \\ 0 & 0 & \eta & \alpha & -\beta \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\mu & 1 & \gamma \end{bmatrix}$$

3.3 $\mathbf{e}[k_0] = [\, 0 \;\; 0 \;\; 0 \;\; 0 \;\; 1 \,]^\top$, coefficients frozen

$$\mathbf{e}[k_0] = [\, 0,\;\; 0,\;\; 0,\;\; 0,\;\; 1 \,]^\top$$

$$\mathbf{e}[k_0+1] = [\, 0,\;\; 0,\;\; 0,\;\; -\beta,\;\; 1 \,]^\top$$

$$\mathbf{e}[k_0+2] = [\, 0,\;\; 0,\;\; q\beta,\;\; -\beta(1+\alpha),\;\; 1 \,]^\top$$

$$\mathbf{e}[k_0+3] = [\, 0,\;\; q\beta,\;\; q\beta(\lambda_c+1+\alpha),\;\; -\beta(1+\alpha+\alpha^2)+\eta q\beta,\;\; 1 \,]^\top$$

$$y_2 = \mathbf{H}_2\mathbf{e}: \quad \gamma,\;\; -\beta+\gamma,\;\; -\beta(1+\alpha) - \mu q\beta + \gamma,$$
$$-\beta(1+\alpha+\alpha^2) + \eta q\beta - \mu q\beta(\lambda_c+1+\alpha) + \gamma,\;\; \ldots$$

$$y_1 = \mathbf{H}_1\mathbf{e}: \quad 0,\;\; 0,\;\; 0,\;\; 0,\;\; q\beta,\;\; q\beta(\lambda_c+1+\alpha),\;\; \ldots$$

3.4 leading order: $\alpha \to 1$, $\eta \to 0$, $\mu \to 0$ (coefficients per step β_i, γ_i, q_i)

$$e_{a_x}[k_0+j] = -\sum_{i=0}^{j-1} \beta_i$$

$$y_2[k_0+j] - \hat{y}_2[k_0+j] = -\sum_{i=0}^{j-1} \beta_i \;+\; \gamma_j$$

$$y_1[k_0+j] - \hat{y}_1[k_0+j] = \sum_{i=0}^{j-3} \lambda_c^{\,j-3-i}\, q_i \sum_{l=0}^{i-1} \beta_l \qquad (j \ge 4;\;\; 0 \text{ for } j \le 3)$$

3.5 the pair (e_{a_x}, e_c), coefficients frozen

$$A_2 = \begin{bmatrix} \alpha & -\beta \\ 0 & 1 \end{bmatrix}$$

$$y_2: \quad C_2 = [\, 1,\; \gamma \,], \qquad \det \begin{bmatrix} C_2 \\ C_2 A_2 \end{bmatrix} = -\beta - \gamma(\alpha - 1) = -\frac{M + \nabla_d x_d\, a' F_{dx}}{\hat{h}^2}$$

$$y_1: \quad C_2 = [\, -q,\; 0 \,], \qquad \det \begin{bmatrix} C_2 \\ C_2 A_2 \end{bmatrix} = -q^2\beta = -\frac{F_{dx}^2\, M}{\hat{h}^2}$$

3.6

$$\det = 0 \quad \Longleftrightarrow \quad M = 0 \text{ and } \nabla_d x_d = 0 \;\; \text{(hold)} \quad \text{or} \quad \hat{h} \to \infty$$

hold: $\beta = \gamma = 0 \;\Rightarrow\; \mathbf{o}_c \equiv \mathbf{0}$ at every $\hat{h}$